Saidmuxammadxon Odilov

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EDUCATION

Central Asian University — Tashkent, Uzbekistan
Bachelor of Science in Computer Science — *Sep 2026 – Jun 2030 (Expected)*

- **Relevant Coursework:** Computer Programming (C++), Calculus, Physics
- **Test Scores:** SAT 1540, IELTS 7.5
- **Codeforces:** 297 practice solutions in Java (github.com/saidkhanodilov/codeforces-practice)

PROJECTS

LearnForFree | *Java, Spring Boot, Spring Security, PostgreSQL, C++17, Docker* — Oct 2026 – Present

- Built a full-stack learning platform where teachers publish courses and quizzes and learners follow learning paths, backed by a 63-endpoint Spring Boot REST API on PostgreSQL with JPA/Hibernate and Flyway
- Secured the API with Spring Security by validating every Supabase JWT (signature, issuer, audience, expiry) and loading learner, teacher and admin roles from the database instead of trusting token claims
- Wrote a C++17 upload inspector that identifies 18 file types by content, not file name, rejecting renamed executables and reading video length from MP4 metadata; its CTest suite runs in the Gradle and Docker builds
- Added daily Web Push reminders and streaks computed in each learner's time zone, using an atomic conditional SQL UPDATE so multiple server instances never send the same reminder twice
- Cached the course catalog in memory instead of reloading every skill, unit and lesson per request, clearing it once an edit commits and using an atomic generation counter so stale loads are never cached
- Wrote 46 JUnit tests, including 15 MockMvc tests covering role-based access and security headers, plus 4 integration suites that run quiz, teacher, reminder and caching workflows on embedded PostgreSQL 17

SimulateUz: 3D Satellite Launch Simulator | *C++17, Java, JNI, OpenGL, JavaScript* — Oct 2026

- Developed a C++17 physics engine that flies a rocket from the launch pad to orbit, modeling J2 gravity, atmospheric drag, pressure-dependent thrust and staging with fixed-step RK4 integration
- Implemented closed-loop ascent guidance and a Hohmann transfer that places a satellite in a 398 × 404 km, 51.61° orbit against a 400 km, 51.6° target, verified by an automated test that flies the full mission
- Rendered the mission in real time with Java and LWJGL (OpenGL 3.3), calling the C++ core through JNI, with up to 1000× time warp and a logarithmic depth buffer spanning 0.5 m to 10 million km
- Ported the physics to a JavaScript and three.js browser game with 5 playable missions; a headless Node.js script flies each one with the autopilot, reaching 400, 600 and 700 km target orbits within 1 km

EXPERIENCE

Freelance Web Developer — During high school
Self-employed — *Uzbekistan*

- Built responsive websites for educational centers in HTML, CSS and Bootstrap, working directly with the owners
- Deployed and maintained the live sites on AWS EC2 instances

LEADERSHIP

AWS Student Builder Group Leader, Central Asian University — 2026 – Present

- Selected by AWS to lead the campus Student Builder Group and launch hands-on cloud workshops for students

Licensee and Organizer, TEDxChelak Youth — Chelak, Uzbekistan

- Obtained an official license from TED to organize a TED-format event for young people in Chelak

SKILLS AND CERTIFICATIONS

Languages: Java (proficient), C++, SQL, JavaScript, HTML/CSS
Frameworks: Spring Boot, Spring MVC, Spring Security (JWT), Spring Data JPA, Hibernate, Flyway, JUnit, Bootstrap
Tools: Git, Maven, Gradle, CMake, Docker, AWS (EC2), Linux, PostgreSQL, Supabase
Certifications: CS50x: Introduction to Computer Science (Harvard University), Responsive Web Design (freeCodeCamp)
Spoken Languages: Uzbek (native), English (fluent)